

Problem 19.1

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$|A| = 1 \cdot (- (1 \times 1 \times 1)) \\ = -1$$

Cofactor method is preferable because the ~~determinant~~ 3×3 matrix we multiply -1 with to get $|A|$ is diagonal thus making it very easy to compute. Row exchange method or using the big formula would take many more steps for the same answer.

Problem 19.2

We know that $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix}$ has determinant 1

Also, by cofactors, the determinant of the second matrix (i.e. the one with det 0) is a sum of products, where one of the products is the value of the $n \times n$ entry multiplied by 1 (i.e. the $n \times n$ entry itself). When the $n \times n$ entry gets reduced by 1, and the sum



$$\begin{aligned} & (\varphi_{200} \psi_{200} \varphi_{1120} +) \varphi_{200} \psi_{200} \psi_{1120} + \\ & \varphi_{200} \psi_{1120} \varphi_{200} \psi_{1120} \psi_{1120} \varphi_{1120} \psi_{1120} -) \psi_{1120} \varphi_{1120} \psi_{1120} - \end{aligned}$$

$$\begin{aligned} & \left(\frac{1}{200} \Phi_{me}^2 \right) \left(\frac{1}{200} \Phi_{me}^2 \right) = \\ & \left(\frac{1}{200} \frac{1}{200} \Phi_{me}^2 \right) \left(\frac{1}{200} \frac{1}{200} \right) + \\ & \left(\frac{1}{me} \frac{1}{200} - \frac{1}{200} \frac{1}{me} \right) \frac{1}{me} \Phi_{me}^2 = \end{aligned}$$

[illegible]

Ques 200 (200) $\Psi_{me} = 9 \times 10^{-30} \Psi_{me} = 0$

$\cos \theta = \frac{1}{2}$
 $\theta = 60^\circ$
 $\therefore \angle A = 60^\circ$
 $\angle B = 60^\circ$
 $\angle C = 60^\circ$

$$\begin{aligned} & (2200 \Phi \text{ m}^2 \text{ g}, 2200 \Phi \text{ m}^2 = \\ & (\Phi_{200} 2200 \Phi \text{ m}^2 \text{ g}) / \Phi_{200} 2200 \Phi \text{ m}^2 \text{ g} \\ & \Phi \text{ m}^2 \Phi \text{ m}^2 \text{ g} \end{aligned}$$

$$2 \text{H}_2\text{O} + 100 \text{O}_2 + 100 \text{H}_2\text{O} + 100 \text{H}_2\text{O} =$$

$$(2 \ln 23 + 64.200) \cdot 0.12^\circ \text{C} =$$

1. The first of the three is the "Secret" -
2. The second is the "Secret" -
3. The third is the "Secret" -

QUALITY